

Problem-Sized Lean Worlds

From a Lean corpus to a short paper, open to outside contribution

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ABSTRACT

Plectis is a public Lean repository organised into problem-sized research worlds, each combining formal mathematics with computations, literature and the questions still open. We describe how a result reaches a short paper through records of its statement, every registered support and the source bytes used in its explanation. A weighted-support criterion for reciprocal Mersenne series provides a worked example of this passage. The same records admit outside contributions and preserve credit when work is revised. In a historical test, nine of ten false edits were rejected, while a false claim about mathematical scope escaped. We locate the resulting division between automated checks and the interpretation required of authors and reviewers.

1 Introduction

A formal theorem can outlive the session in which it was proved, yet its hypotheses, motivation and relation to an open question may be lost when it is turned into prose. We organise the research around a persistent problem world: the question, its sources, the formal and computational work already done, and the precise obligations that survive. Figure 1 follows one claim from that world to a paper and back through a contribution.

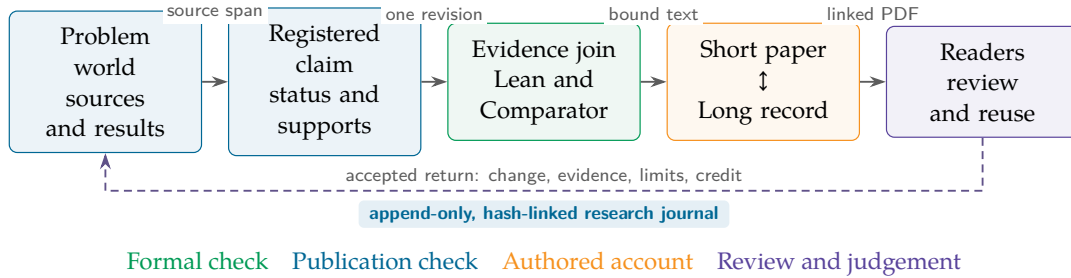


Figure 1: The life of one claim: arrows name the objects exchanged, the short and long accounts remain linked, and colours distinguish kinds of responsibility throughout the paper.

Erdős Problems supplies the questions and their public discussion, while Lean and mathlib supply the formal language and library [1, 10, 75]. Blueprints already connect exposition to declarations [34, 73], and persistent problem records and open agent collaboration have antecedents in ETP and Prove2Me [74, 30]. Our publication tools add a check of the complete registered support set and bind the surrounding prose, including explanations and limitations, to specified source bytes. The following sections describe the world, trace a weighted-support theorem through those records, and separate the checks from the judgements made when contributions enter the corpus.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

2 A problem-sized Lean world

A world has one endpoint question and many possible contributions: a proof, a counterexample, an exact experiment, a corrected reference or an explanation of a failed method. We expose four views of the existing mathematics before asking a reader to compile Lean (Figure 2). The declaration atlas locates every declaration in its inventory; the dependency index records exact edges for its loaded roots. The semantic graph adds authored interpretation, and the claim registry selects assertions with their status and open obligations.

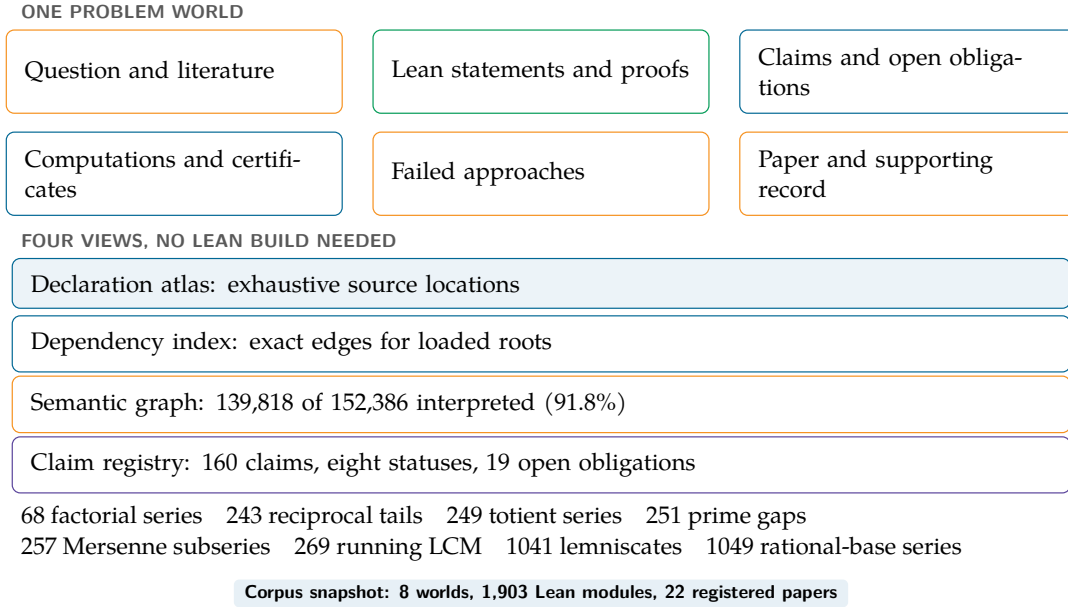


Figure 2: A world contains the evidence and unresolved work for one question; the counts describe the whole recorded corpus, and exhaustive declaration locations do not imply exhaustive mathematical interpretation.

A query can therefore start with a paper claim and return its source location or dependency neighbourhood without treating an authored summary as a proof. An interpretation record may supply context rather than a proof-level explanation; these counts do not record an individual human reading of every declaration. After locating the relevant source, a contributor can request a focused build and inspect its receipt rather than infer compilation from navigation.

3 One theorem, end to end

For an infinite set A of positive integers and an integer $b \geq 2$, write $X_A(b) = \sum_{a \in A} (b^a - 1)^{-1}$. Given a finite nonempty set P of primes, let $h_P(a) = \prod_{p \in P} p^{v_p(a)}$ be the part of a supported on those primes. The weighted-support theorem of the Problem 257 paper states that, if

$$W_{2,P}(A) := \sum_{a \in A} \frac{h_P(a)}{a(2^{h_P(a)} - 1)} < \infty, \quad (1)$$

then $X_B(b)$ is irrational for every integer $b \geq 2$ and every infinite $B \subseteq A$ [2, Theorem 1.2]. The paper also gives a fixed-base version with 2 replaced by b in the hypothesis; the recorded Lean support for both clauses is shown in Figure 3. The arbitrary infinite-support question in base two remains open.

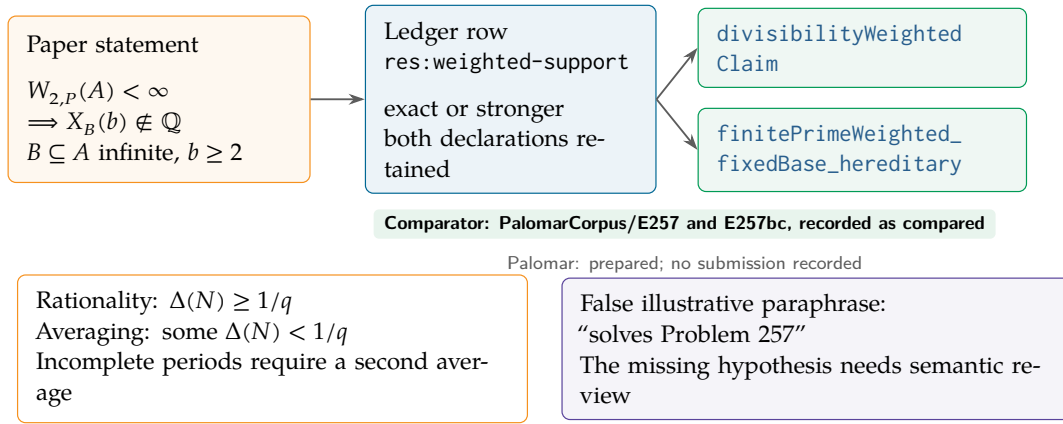


Figure 3: The weighted-support statement is joined to both registered declarations; the false paraphrase at lower right illustrates a meaning error and is distinct from the historical edit 8 in Table 1.

Erdős had already stated that $\sum_{a \in A} 1/a < \infty$ suffices for this all-base irrationality conclusion [3, p. 222]. The corresponding formal declaration is `irrational_erdosSupportSeries_of_summable_reciprocal`, with Comparator entry PalomarCorpus/E257av recorded as compared. The source paper credits Duverney and Tachiya for an arithmetic-progression averaging argument of the kind used to select a small positive displacement [4, Section 2, (2.3)–(2.9)].

To see what the weighted hypothesis adds, consider the source paper's calculation

$$A_* = \{2^k m : k \geq 1, m \text{ odd}, 1 \leq m \leq 2^{2^k}\}, \quad P = \{2\}.$$

Put $S_r = \sum_{1 \leq m \leq 2^r, m \text{ odd}} 1/m$; grouping the odd integers into intervals $[2^j, 2^{j+1})$ gives $r/4 \leq S_r \leq r$ for integers $r \geq 2$. The disjoint layer with $v_2(a) = k$ therefore contributes at least $2^{-k} S_{2^k} \geq 1/4$ to the reciprocal sum, whereas its weighted contribution is at most

$$\frac{S_{2^k}}{2^{2^k} - 1} \leq \frac{2^k}{2^{2^k} - 1} \leq 2^{1-k}.$$

Thus $\sum_{a \in A_*} 1/a$ diverges but $W_{2,\{2\}}(A_*)$ converges; this calculation is supplied in the paper's example and has no separate Lean-ledger row [2, calculation following the weighted proof].

The proof uses the nonnegative displacement

$$\Delta_{b,A}(N) = \sum_{a \in A} \frac{b^{N \bmod a} - 1}{b^a - 1} = (b^N - 1)X_A(b) - J_{b,A}(N), \quad N \geq 1, \quad (2)$$

where the finite sum $J_{b,A}(N) = \sum_{a \in A, a \leq N} \sum_{j=1}^{\lfloor N/a \rfloor} b^{N-ja}$ is an integer. If $X_A(b) = p/q$ with $q > 0$, then $\Delta_{b,A}(N) > 0$, because A contains an exponent larger than N , and integrality in (2) gives $\Delta_{b,A}(N) \geq 1/q$.

We seek a multiple $N = tQ$ at which this displacement is smaller. A finite part of A accounts for all but an arbitrarily small weighted tail, and taking Q divisible by that finite part makes its displacement terms vanish. Enlarging Q by suitable powers of the primes in P controls the orbit averages of the other terms through $\gcd(a, Q)$.

The difficulty is that a finite average usually ends partway through an orbit: bounds for those incomplete periods need not sum over all exponents. The source proof takes a second average over lengths 2^j , with $M \leq j < 2M$, choosing M large relative to the polynomial size of Q and small relative to the exponential orbit saving. Both boundary errors then tend to zero, so one

term contradicts the rational lower bound; the weighted sum decreases on passing to $B \subseteq A$, and $b^{h_P(a)} - 1 \geq 2^{h_P(a)} - 1$ transfers the binary hypothesis to every $b \geq 2$ [2, Section 2].

4 Proof and publication checks

We bind a paper statement to an identified source revision and compare its complete registered support set, including named inputs, with the evidence supplied for it. A record for one supporting lemma cannot stand for a result that requires several, and a changed source or statement requires the corresponding records to be checked again. Figure 4 distinguishes the relations checked by the kernel, Comparator and publication tools.

Relation	Lean	Comparator	Tools	Author	Review
Formal statement and proof	●				
Challenge and formal statement		●			
Sentence and declared source bytes			●		
Sentence and mathematical meaning				○	○
Result and motivating question				○	○
Novelty, significance and acceptance					○

Figure 4: Filled circles denote machine checks and open circles denote judgments; a source binding concerns declared bytes, while the relation between a sentence and its mathematical meaning remains a separate review question.

Lean verifies that a proof establishes the formal statement written in the source; it does not verify whether that statement captures the intended mathematics or whether the paper describes it well. Comparator adds a separately stated challenge and a check of its selected Lean implementations under the permitted axioms [18, 19].

For prose, we retain the exact sentence and source span together with an evidence class and an authored warrant explaining the connection. The sentence checker compares those digests and rejects missing bindings or unbound prose. An author may revise a sentence and its warrant together while keeping the hashes current, so agreement of the bytes still leaves the meaning to be reviewed. Dropping the support condition from an abstract then asserts more than the theorem proves, even though its Lean links remain intact.

An atlas entry with a matching signature and a reachable target attests to that source projection; it does not establish that any particular continuous-integration run compiled the module. The notary interface derives probe verdicts from the pinned Lean process, so a caller cannot submit its own verdict through that interface. This interface rule does not prevent an actor with filesystem access from changing the implementation or stored record. The protocol distinguishes contributor and reviewer roles but does not technically force a second independent mathematician.

5 From a Lean corpus to a paper

We use the dependency graph to locate the argument and then decide which contribution deserves the short paper, taking account of its natural generality and antecedents. The author supplies the proof strategy and explains the difficult step beside the registered statement. The long record provides the complete argument and subsidiary cases, and an accepted correspondence identifies the passage supporting a particular short-paper claim. The assembler combines

these authored parts and preserves their cross-references, without supplying a new mathematical argument.

5.1 Source-frozen editorial refinement

We first prepare a dossier from the current corpus: candidate results, their antecedents, the difficult proof transitions and the exact questions left unresolved. The author chooses the principal argument; the dossier supplies source locations and evidence, while the native manuscript checker audits source consistency and the specified manuscript rules. For a further revision, we freeze the selected short paper and long record together with their input closure, coverage ledger and native audit rules in a packet whose manifest identifies the exact source bytes. The writer may be an advisory model returning a proposed revision, or a tool-enabled agent doing the same authoring work in a checkout; either arrangement leaves mathematical acceptance with the integrating reviewer.

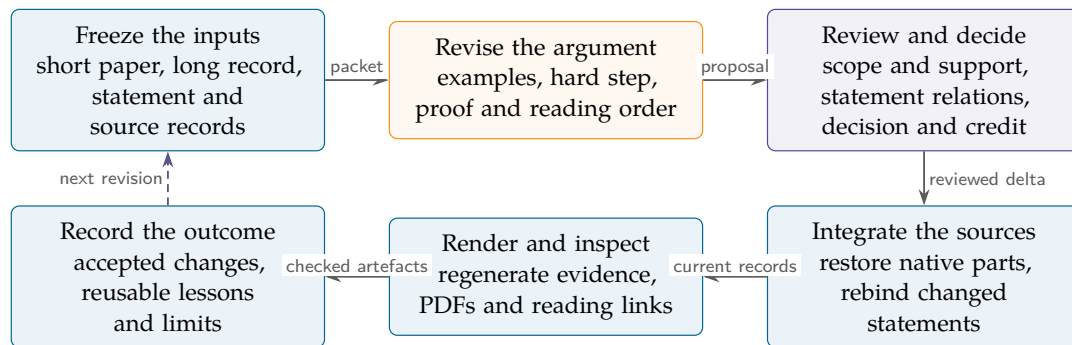


Figure 5: The writer returns a proposal against frozen sources. Review precedes integration; rendering and inspection precede the recorded outcome and next revision. Advisory and tool-enabled writers share this acceptance boundary.

The native return checker verifies packet identity, permitted edit paths, statement-change declarations and preservation of labels, citations and bibliography, then reports the manuscript audit on a temporary source overlay. It neither applies the proposal nor executes returned programs, and a declared statement change remains a mathematical review obligation. The reviewer checks changed hypotheses and implication directions against the whole statement and its registered declarations, retaining an ordinary argument with its actual evidence class when formalisation is incomplete. After that review, we restore the long record's authored parts, regenerate evidence and navigation, compile the manuscripts and inspect the resulting figures and proof links. A useful returned writing rule enters the next revision as a candidate with its example and limits; adoption requires a reviewed improvement to the owning writing contract. Figure 5 describes this reproducible workflow, whose packet and audit stages are implemented by `short_paper_writer.py`; it supplies no measurement of reader benefit or autonomous discovery.

The relation to the motivating problem must also survive compression: the Problem 269 record distinguishes distinct running-LCM heights from the heights counted with multiplicity in the original question [27]. A result about the former is not automatically a result about the latter. When a statement is sharpened, merged or moved, its complete support set and any named input must move with it, and the statement relation needs renewed review. A new arrangement of a proof does not make its ingredients new.

6 An open route for contributions

A contributor may send a mathematical idea or reference without a clone, or return a patch from a recorded public commit. For executable work the return identifies the changed files, commands, results, resource use and surviving limitations, with the people and tools responsible. Figure 6 separates the proposed change from replay, review and adoption. We first reproduce an older branch at its original base and then reconcile it with current main, assigning separate credit to any substantive integration repair.

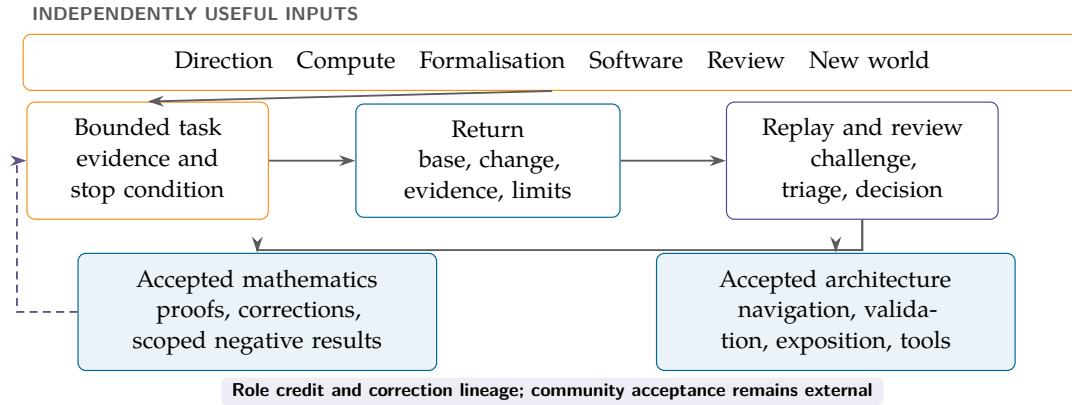


Figure 6: The contribution route accepts mathematical and architectural work under their own evidence classes; the feedback arrow records an adopted change and does not certify independent review or increased discovery rate.

The research journal records returns and dispositions in an append-only, hash-linked history, while a correction may withdraw something previously endorsed. Discovery investigates a question, while stewardship compares the return with prior work, groups related results, reconciles affected papers and selects the next question. The proposed coupling is event-driven: a stable result or changed appraisal prompts work, whereas an unchanged repository supplies no reason for another run. The journal establishes what happened; judging what was learned requires the comparison with earlier work.

Generalising a local result requires a separate literature and library comparison, a proof of the proposed extension, and a second useful instance before an expert takes it to an upstream project [100]. That route is not automated and has produced no reported Mathlib contribution in the supplied record. We distinguish credit for an idea from credit for its implementation, formalisation or exposition, using CRediT as a role vocabulary rather than an authorship decision [104]. Later use adds an attributed dependency or correction, with neither a scalar impact score nor a financial contract inferred from that entry. A new world likewise needs a sourced question, explicit status, inspection routes and a review owner, which must still be assembled in several steps.

7 Recorded observations

The frozen inventory contains 693 paper occurrences, of which 644 have exact or specialising Lean support; 23 depend on a named input and 26 have no recorded Lean support. Of the 644 exact-or-specialising occurrences, 620 are recorded as compared and 24 as queued. These are classifications of statement occurrences, with the named-input and unsupported entries exposed separately from the fully covered entries.

The historical publication-evidence record reports that nine of the ten deliberately false edits were rejected and one escaped. The edits were authored by the checker’s author, the original run logs were not retained, and the other nine edits were not rerun against the extended checklist. The edits were applied separately, with the baseline restored between trials. The escaped edit changed a description of an unresolved step from “does not supply” to “completes”, and the publication checks accepted it.

Edit	Change	First detector	Outcome
edit 1	Conditional result labelled proved	Projection freshness	rejected
edit 2	Open-boundary clause deleted	Boundary wording	rejected
edit 3	Declaration coordinate moved	Source coordinates	rejected
edit 4	Paper source link retargeted	Source coordinates	rejected
edit 5	Claim assigned to two families	Registry structure	rejected
edit 6	Module count hand-edited	Projection freshness	rejected
edit 7	native_decide introduced	Proof-trust policy	rejected
edit 8	“does not supply” changed to “completes”	None	escaped
edit 9	Open-problem title mangled	Paper anchoring	rejected
edit 10	Orientation byte budget exceeded	Byte budget	rejected

Table 1: The ten historical edits and their recorded first detectors; “proof-trust policy” refers to this repository’s permitted proof mechanisms, not to a general unsoundness claim about native_decide.

After the escape, the follow-up checked the intact baseline and that particular edit against an added check, leaving the other edits untested under the repair and supplying no post-repair ten-of-ten result. The post-repair witness accepts the current README and rejects a test copy containing the false clause. The sample comes from one corpus and one author’s selection of errors, with neither a manual-review control nor an ordinary continuous-integration comparator. The study locates a coverage boundary. Its nine rejections do not estimate how reliable the checker is.

One architecture return was accepted on 23 September 2026 after focused checks of the ordinary-language return route and an architecture-packaging case. The contributor and maintainer-operated reviewer were the same agent; neither the receipt nor this paper claims an independent outside clone replay. As of 14 September 2026, no completed external cold-clone use had been recorded.

For readers, the proposed comparison holds the mathematical task fixed and contrasts an ordinary repository, an information-equivalent static briefing and the navigation tools. It would record understanding, severe errors and the costs of preparation, use, review and repair, including missing returns, timeouts and disagreements. No independent writing comparison, blind grading run or cold-reader experiment is reported here, and no comparative reader result is reported. General reliability, reader understanding and mathematical transfer remain unmeasured.

8 Related work

Figure 7 locates the inherited components and the wider literature retained from the three source accounts. Prove2Me’s first arXiv version appeared on 28 August 2026, before the first public version of this paper, and already described public agent contributions, task decomposition,

reusable statements and credit for reuse [30]. Its hosted verification service and this repository-based record provide different places for those activities; we claim neither priority for persistent research records nor a measured advantage over these systems. Polymath, BOINC and GIMPS supply precedents for small mathematical contributions and donated compute [99, 97, 98], while the Carleson blueprint shows formalisation organised into publicly claimable tasks [87].

The totient record refines Coons's non- k -regularity antecedent and distinguishes Martin's separation theorem from its own Chinese-remainder and Dirichlet argument traced to Yazdani [12, 13, 14, 15]. The running-LCM record retains Fan's public priority for the two-prime case despite an independently reported derivation, and credits its Hecke–Mahler antecedents separately [27, 28, 25, 26]. For Problem 1041, the public degree-seven construction is ani's; the recorded Lean result refutes the exact Formal Conjectures statement, while correspondence with the 1958 curve-length wording remains unreviewed [22, 76, 23, 21, 20]. Appendix A names its four recorded Lean declarations; Comparator is pending. The other seven targets remain unresolved here.

9 Limits

This paper reports no new Lean or Comparator execution, comparative formalisation result or autonomous-optimisation experiment. Transfer to unseen mathematics and understanding by independent human readers are unresolved. The record supplies no controlled evidence that the contribution machinery improves the conversion from compute to mathematics or increases discovery rate. It also supplies no evidence that a new project can adopt the architecture cheaply or that an agent using it outperforms one using a good README. Full environment scans and initial builds still have costs, and an open corpus cannot establish that a model has never encountered its material.

The project remains maintainer-centred, with scarce review capacity and contestable credit; openness alone does not equalise resources or resolve conflicts and appeals. Broad mathematical acceptance is exogenous to this repository and cannot be granted by its maintainer. Returned code is untrusted, and expensive or privileged continuous-integration jobs must not execute fork code with repository secrets [105]. A volunteer-compute service and extensions to physical science remain proposals requiring their own validation, resource and safety governance. The present contribution is an inspectable route through the records: a reader can recover the hypothesis, argument and support of a stated result, and identify where a source check ends and mathematical review begins.

Source family	Role in the three source accounts
Formal foundations	Lean and mathlib; library maintenance and contributor practice [10, 75, 92, 100]
Source and statement checks	Comparator, kernel replay, vacuity, linters and property tests [18, 19, 51, 42, 50]
Formal-to-prose tools	Blueprints, source documentation, translation and refactoring [34, 73, 36, 35, 60, 69]
Retrieval and interaction	Premises, proof states and exact declaration neighbourhoods [38, 88, 90, 91, 89, 93]
Proof automation	Premise selection and tactic search [48, 49, 58]
Library structure	Declaration networks and repository knowledge graphs [44, 45, 95]
Assumptions and reuse	Proof mining, generalisation, lemma mining and theory exploration [40, 41, 52, 53]
Growing proof knowledge	Library learning, evolving procedures and candidate statements [54, 55, 56, 57, 61, 83]
Research agents	Executable objectives, proof cost and evolving tasks [77, 78, 79, 29, 24, 59]
Agent collaboration	Bounties, planner/worker state, retrieval and compiler feedback [80, 81, 82, 30, 43, 74]
Evaluation	Research budgets, proof engineering and documentation tasks [84, 70, 71, 72, 106, 102, 103]
Validity and intended meaning	Research case studies, interactive proving and public accounting [17, 31, 39, 32]
Public infrastructure	Problem sources, formal conjectures, verification and collaboration [1, 76, 101, 87, 99, 97, 98, 104, 105]
Writing and maintenance	Mathematical exposition, build dependencies and provenance [85, 68, 47, 46, 94]
Understanding and reuse	Explanatory generality, digestion and research possibilities [5, 6, 9, 86, 7, 62, 96]
Different community views	Declarations, dissent, benchmarks, research infrastructure and pace [8, 63, 64, 65, 33, 66, 67]
Mathematical antecedents	Weighted support, totient ranks and arithmetic separation [2, 3, 4, 12, 13, 14, 15]
Other mathematical records	Running LCM, lemniscates and cross-problem connections [27, 28, 25, 26, 11, 37, 21, 20, 22, 23, 16]

Figure 7: Sources grouped by the roles for which the original accounts cite them, rather than by a performance ranking; the accompanying mathematical attribution distinguishes original results, refinements and independent rediscovery.

A Inspection and reproduction

This revision builds on the frozen manuscript and evidence packet recorded at [ced96e58b337](#), a local integration commit rather than public main. Source links in this revision point to [7308896010e6b527d283dfdd0687a717a6183fb6](#); the Comparator links below identify their separate recorded runs. The two companion manuscripts remain unchanged provenance inputs for this consolidation.

The weighted row in [docs/paper_lean_coverage.json](#) names `divisibilityWeightedClaim` and `finitePrimeWeighted_fixedBase_hereditary`, in namespace `ErdosProblems.Erdos257.PaperCompleteR8`. Its Comparator records are [run 35544127144](#) and [run 35624228171](#); the latter also records `PalomarCorpus/E257av` for the reciprocal antecedent. Both runs are recorded as compared; Palomar is prepared and not submitted.

The four declarations supporting the reported 1041 refutation have namespace `Erdos1041.Counterexample`: [erdos1041_counterexample](#), and the [HausdorffLength.lean](#) declarations `erdos1041_counterexample_hausdorff`, `erdos1041_hausdorff_negation` and `erdos1041_hausdorff_answer_false`. The coverage row gives exact-or-stronger Lean support, with Comparator pending.

In a complete checkout, `python3 scripts/systems_paper_evidence.py` checks this paper's source bindings, and `python3 scripts/build_systems_paper_counts.py` checks the generated inventory. The record at [docs/publication_evidence.json](#) holds the historical matrix and the limited follow-up, while [docs/publication_contract.json](#) declares the publication checks. Formal entry points include [Erdos249257.lean](#) and [.github/workflows/lean.yml](#); the curated assertions are in [docs/claims.json](#), and [scripts/check_release.py](#) runs the release gate. The frozen packet omits the Lean checkout and several full-repository inputs, so source-binding validation here is not a replay of those broader checks.

The protected snapshot partitions 693 occurrences into 502 exact, 142 specialising, 23 named-input and 26 unsupported-by-Lean entries. The journal summary at head [d3ad85269386](#) records 7 rounds (2 sealed), 25 returns (18 admitted and 7 unreviewed); earlier private-history rounds are not a complete public journal. The semantic counts in Figure 2 are the supplied interpretation inventory, not a count of individually reviewed proofs. Public reading begins with [the reading guide](#); [CONTRIBUTING.md](#) describes contribution channels under the existing [credit policy](#).

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